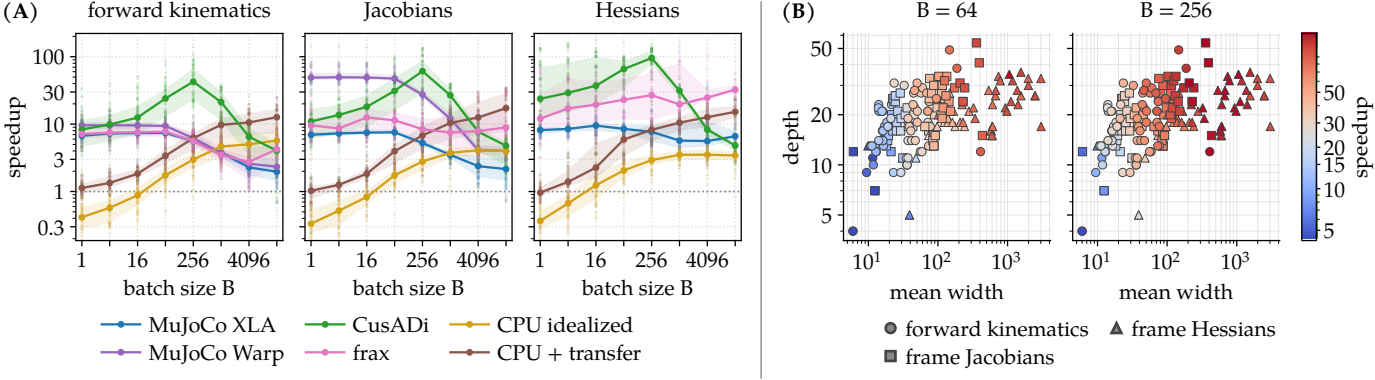

(A)
forward kinematics
Jacobians
Hessians
speedup
100
30
10
3
1
0.3
1
16
256
4096
batch size B
MuJoCo XLA
MuJoCo Warp
CusADi
frax
CPU idealized
CPU + transfer
(B)
B = 64
B = 256
depth
50
20
10
5
$10^1$
$10^2$
$10^3$
mean width
speedup
50
30
20
15
10
5
forward kinematics
frame Hessians
frame Jacobians